

# Hierarchy of Numbering Systems

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- **Natural Numbers** ( $\mathbb{N}$ ):

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

(Some definitions also include 0:  $\{0, 1, 2, 3, \dots\}$ ).

- **Whole Numbers**:

$$\{0, 1, 2, 3, \dots\}$$

- **Integers** ( $\mathbb{Z}$ ):

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

- **Rational Numbers** ( $\mathbb{Q}$ ):

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0 \right\}$$

Rational numbers include all numbers that can be expressed as a ratio of two integers.

- **Irrational Numbers**:

$$\{\sqrt{2}, \pi, e, \dots\}$$

Irrational numbers are numbers that cannot be expressed as a ratio of two integers, with non-repeating and non-terminating decimal expansions.

- **Real Numbers** ( $\mathbb{R}$ ):

$$\mathbb{R} = \text{Rational Numbers} \cup \text{Irrational Numbers}$$

The real numbers include both rational and irrational numbers and represent every point on the continuous number line.

- **Imaginary Numbers** ( $\mathbb{I}$ ):

$$\mathbb{I} = \{i \cdot b : i = \sqrt{-1}, b \in \mathbb{R}\}$$

Imaginary numbers involve the square root of negative numbers, where  $i = \sqrt{-1}$ .

- **Complex Numbers ( $\mathbb{C}$ ):**

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}, i = \sqrt{-1}\}$$

Complex numbers combine real and imaginary numbers in the form  $a + bi$ .